

Comparing Deep Reinforcement Learning Algorithms for Continuous Locomotion Control in MuJoCo

Ahmad Hassan

Khoury College of Computer Sciences, Northeastern University

NUID: 002560629

hassan.ahmad@northeastern.edu

Abstract

We present an empirical comparison of three deep reinforcement learning algorithms—Twin Delayed DDPG (TD3), Soft Actor-Critic (SAC), and Proximal Policy Optimization (PPO)—on three MuJoCo locomotion benchmarks of increasing difficulty: HalfCheetah-v5, Hopper-v5, and Walker2d-v5. Unlike prior work that relies on off-the-shelf libraries, every algorithm is implemented from scratch in PyTorch following the canonical papers and the CleanRL reference conventions, which surfaces the implementation details that dominate performance in practice. Each of the nine algorithm–environment pairs is run with five random seeds for one million environment steps, giving 45 total training runs and a further 54 runs for a hyperparameter sensitivity study on HalfCheetah. The off-policy actor–critic methods (TD3 and SAC) dominate on HalfCheetah, reaching roughly ten times the return of on-policy PPO, but the gap closes dramatically on Hopper—where PPO is actually competitive—and reverses on Walker2d, where SAC leads. Sensitivity experiments reveal that SAC’s auto-tuned entropy coefficient is essential (reverting to a fixed $\alpha = 1.0$ loses roughly 28% of final return), that PPO collapses catastrophically once the clip range exceeds 0.2, and that TD3 tolerates a wide range of exploration noise scales.

Introduction

Continuous control locomotion is a canonical proving ground for deep reinforcement learning algorithms. It combines high-dimensional observations and actions with contact-rich dynamics that are hard to model analytically, so the same simulation infrastructure underpins research on robot walking, prosthesis design, and neuromechanical modelling of human gait (Song et al. 2021). Among the many algorithms that have been proposed for this setting, three have emerged as the de facto baselines: PPO (Schulman et al. 2017), SAC (Haarnoja et al. 2018), and TD3 (Fujimoto, van Hoof, and Meger 2018). They occupy fundamentally different points in the design space—on-policy versus off-policy, stochastic versus deterministic, entropy-regularised versus standard—and each is widely recommended as a “safe default” in its own community.

Despite their ubiquity, direct comparisons across these three algorithms are often hard to interpret. Seemingly mi-

nor implementation choices—network initialisation, reward scaling, whether advantages are normalised at the minibatch level—can change conclusions about which algorithm is better, and results reported on one MuJoCo environment often fail to generalise to another. This is partly why every serious benchmark paper re-runs all baselines under a single codebase rather than copying numbers from elsewhere.

This project takes that lesson seriously. Rather than use an off-the-shelf library such as Stable-Baselines3 (Raffin et al. 2021)—which was the original plan in our project proposal—we implement TD3, SAC, and PPO from scratch in PyTorch, following the original papers and the CleanRL (Huang et al. 2022) single-file reference as a correctness check. This pedagogical choice made the project substantially more work, but it forced us to confront the small implementation details that turn out to dominate results in practice: orthogonal initialisation with a near-zero actor output gain, per-minibatch advantage normalisation and clipped value loss for PPO, automatic entropy tuning for SAC, and delayed policy updates with target policy smoothing for TD3.

The contributions of this report are (i) a clean, fully open implementation of TD3, SAC, and PPO in a single codebase; (ii) learning curves on three MuJoCo locomotion environments averaged over five seeds, against a uniform random-policy lower bound; (iii) sample-efficiency and final-performance comparisons; and (iv) a hyperparameter sensitivity study on HalfCheetah covering the learning rate, TD3 exploration noise, SAC entropy coefficient, and PPO clip range.

Related Work

Reproducibility has been a persistent concern in deep reinforcement learning: benchmarking studies have repeatedly found that conclusions about which algorithm dominates on one MuJoCo environment often fail to carry over to another, and that results depend heavily on codebase and random-seed choices (Ahmad et al. 2023). The response in the community has been to re-run baselines under a single controlled codebase rather than trust cross-paper comparisons.

A complementary strand of work targets the implementation details themselves. CleanRL (Huang et al. 2022) provides single-file reference implementations of all the major deep-RL algorithms and explicitly documents the small

design choices—orthogonal initialisation, per-minibatch advantage normalisation, value-loss clipping, learning-rate annealing, reward normalisation—that dominate continuous-control PPO performance in practice. Our work follows the same “re-implement and verify” methodology on a smaller scale: every algorithm is written from scratch and cross-checked against the CleanRL reference for PPO and the canonical papers for TD3 and SAC.

The alternative is a library-style interface such as Stable-Baselines3 (Raffin et al. 2021), which bundles validated implementations behind a uniform API. This is convenient but hides the exact details that we wanted to surface; we deliberately did not use it.

Background

We briefly recap the three algorithms, highlighting the mechanisms that most affect behaviour in practice.

PPO. On-policy. At each iteration the policy collects a rollout of T steps, estimates advantages with Generalised Advantage Estimation (Schulman et al. 2017), then runs several epochs of stochastic gradient ascent on the clipped surrogate objective $L^{CLIP}(\theta) = \mathbb{E}[\min(r_\theta \hat{A}, \text{clip}(r_\theta, 1-\epsilon, 1+\epsilon)\hat{A})]$, where r_θ is the probability ratio between the new and old policies. The clip prevents destructive updates at the cost of sample efficiency.

SAC. Off-policy. Maximises a maximum-entropy objective $\mathbb{E}[\sum_t r_t + \alpha \mathcal{H}(\pi(\cdot|s_t))]$ using a stochastic squashed-Gaussian actor and twin soft Q-networks. The entropy temperature α can either be fixed or—more robustly—learned online so that the average entropy tracks a target $\mathcal{H}_{\text{target}} = -\dim(\mathcal{A})$.

TD3. Off-policy, deterministic. Extends DDPG with three decisive tweaks: two Q-networks with the *minimum* of their predictions used as the Bellman target (to curb overestimation); target-policy smoothing (Gaussian noise is added to the target action before evaluating the target Q); and delayed policy updates (the actor is updated every d -th critic step). Exploration at data-collection time uses Gaussian action noise with a fixed standard deviation.

Methods

All three algorithms share a common codebase. The MLP trunks are identical (two hidden layers of 256 units) so comparisons reflect algorithmic rather than architectural differences. The heads and activation functions, however, differ where the algorithm demands it.

TD3 networks. A deterministic actor passes the hidden output through tanh and scales by `max_action`. Two independent Q-networks concatenate (s, a) at the input layer. Hidden activations are ReLU. Target networks for actor and critics are maintained by Polyak averaging with $\tau = 5 \times 10^{-3}$. Training follows Algorithm 1 of Fujimoto, van Hoof, and Meger (2018): target noise $\sigma_{\text{tgt}} = 0.2$ clipped to ± 0.5 , exploration noise $\sigma_{\text{exp}} = 0.1$, policy delay $d = 2$. After a 25k-step uniform random warm-up the agent updates on a minibatch of 256 transitions per environment step, sampled from a replay buffer of size 10^6 .

SAC networks. A squashed-Gaussian actor emits $(\mu, \log \sigma)$, samples $u \sim \mathcal{N}(\mu, \sigma)$ with the reparameterisation trick, and returns $a = \tanh(u)$; the tanh correction $-\sum_i \log(1 - a_i^2 + \epsilon)$ is subtracted from the log-probability to keep the policy gradient unbiased. Twin soft Q-networks and their targets are updated at every environment step after a 5k-step warm-up (shorter than TD3’s because the stochastic policy explores on its own). The entropy temperature α is learned online by gradient descent on the loss $\mathcal{L}(\alpha) = -\alpha(\log \pi(a|s) + \mathcal{H}_{\text{target}})$ with $\mathcal{H}_{\text{target}} = -\dim(\mathcal{A})$.

PPO networks and wrappers. PPO is where implementation details bite hardest, so we followed the CleanRL (Huang et al. 2022) continuous-control reference closely. The actor and critic are separate MLPs with tanh hidden activations, orthogonally initialised with gain $\sqrt{2}$; the actor output layer uses a small gain of 0.01 (so initial action means are near zero and the policy starts with low control cost) while the critic output uses gain 1.0. The action log-standard-deviation is a single state-independent learnable parameter initialised to zero. On the rollout side we wrap the environment with `ClipAction`, then `NormalizeReward` (a running discounted-variance rescaling with $\gamma = 0.99$), then a ± 10 reward clip. During updates we normalise advantages *per minibatch*, apply the symmetric clipped value loss of CleanRL, and linearly anneal the learning rate to zero over the 1M-step budget. These details are not cosmetic—removing any one of them degraded HalfCheetah performance substantially in preliminary runs.

A note on the original proposal. Our project proposal planned to use Stable-Baselines3 (Raffin et al. 2021) for all three algorithms to “ensure a fair comparison”. After the proposal was reviewed we decided to implement every algorithm from scratch instead. The motivation was pedagogical: using a black-box library would make it possible to get results without ever internalising *why* PPO needs clipped value losses, or *how* automatic entropy tuning actually changes the SAC update. Re-implementing the three algorithms end-to-end forced exactly those questions into view, and the additional implementation effort was more than offset by the diagnostic insight it gave us into the failure modes of PPO in particular.

Experimental Setup

Environments. We use three Gymnasium (Towers et al. 2024) MuJoCo (Todorov, Erez, and Tassa 2012; Farama Foundation 2024) locomotion tasks: HalfCheetah-v5, Hopper-v5, and Walker2d-v5. All three are continuous-state, continuous-action MDPs with actions bounded in $[-1, 1]$. HalfCheetah has 17 observation dimensions and 6 action dimensions, cannot fall, and rewards forward velocity minus a small control cost. Hopper (11-dim state, 3-dim action) and Walker2d (17-dim state, 6-dim action) add a healthiness reward and terminate episodes when the robot’s torso drifts outside a stable region. The three environments form a natural difficulty progression: HalfCheetah rewards sheer forward-thrust acceleration, Hopper requires balancing on one leg, and Walker2d requires coordinated bipedal gait.

Hyperparameter	TD3	SAC	PPO
Hidden layers	256, 256	256, 256	256, 256
Hidden activation	ReLU	ReLU	Tanh
Optimiser	Adam	Adam	Adam
Learning rate	3×10^{-4}	3×10^{-4}	1×10^{-4}
Discount γ	0.99	0.99	0.99
Polyak τ	0.005	0.005	—
Batch size	256	256	64
Buffer / rollout	10^6	10^6	2048
Warm-up (steps)	25,000	5,000	—
Target noise / clip	0.2 / 0.5	—	—
Exploration noise	0.1	—	—
Policy delay	2	—	—
Initial α	—	1.0 (auto)	—
GAE λ	—	—	0.95
Clip ϵ	—	—	0.2
Value-loss coef.	—	—	0.5
Entropy bonus	—	—	0.0
Update epochs	—	—	10
Max grad norm	—	—	0.5
LR annealing	—	—	linear

Table 1: Hyperparameters used for all main experiments. Values follow the canonical papers (Fujimoto, van Hoof, and Meger 2018; Haarnoja et al. 2018; Schulman et al. 2017) and the CleanRL (Huang et al. 2022) PPO reference.

Protocol. Each algorithm runs for 10^6 environment steps on each environment, with five random seeds per (algorithm, environment) pair. Every 10^4 steps we pause training and run ten deterministic evaluation episodes (stochastic algorithms use the mean action) on a fresh copy of the environment, seeded differently from the training env. All plots below use these evaluation returns, not training-time episode returns; this avoids conflating exploration noise with policy quality. As a lower bound we also record a uniform random-action baseline under the same protocol.

Hyperparameters. We use the canonical published hyperparameters for all three algorithms; see Table 1 for the full list. Unless otherwise stated these are the values used for every main run. No environment-specific tuning was performed—one config per algorithm was reused across all three environments—since the goal is to compare out-of-the-box behaviour rather than squeeze the last few percentage points from each task.

Results

Main Results

Figure 1 and Table 2 summarise the final evaluation returns for every (algorithm, environment) pair. The learning curves for the three environments are shown in Figure 2.

HalfCheetah. This is the environment where the continuous-control “textbook” story holds most clearly. TD3 and SAC both reach around 10,000 evaluation return and are visually indistinguishable on the learning curves after the first 300k steps; they both exceed PPO’s final return by more than an order of magnitude (861 ± 276). TD3 is slightly ahead in the mean but SAC has higher

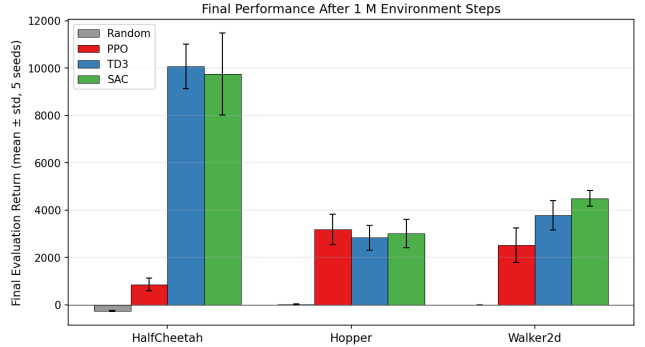


Figure 1: Final evaluation return after 1M environment steps. Bars show the mean across five seeds; error bars are one standard deviation. The random-action baseline is essentially zero on all three environments.

Algorithm	HalfCheetah	Hopper	Walker2d
Random	-264 ± 16	22 ± 9	2 ± 1
PPO	861 ± 276	3183 ± 636	2527 ± 730
TD3	10070 ± 936	2833 ± 516	3784 ± 625
SAC	9747 ± 1727	3015 ± 605	4499 ± 325

Table 2: Final evaluation return (mean ± std over 5 seeds) at 1M environment steps. Best value per environment in bold where a single method is clearly ahead.

variance across seeds, so the gap between the two is not statistically meaningful given five seeds. SAC’s higher variance is driven largely by one seed out of five that converged to 6,746—roughly 35% below the other four seeds—indicating that even with automatic entropy tuning, SAC can occasionally settle into a suboptimal basin on this environment. We note that our PPO score on HalfCheetah sits below typical published figures in the 1500–3000 range at 1M steps; we suspect the NormalizeReward wrapper combined with the ± 10 reward clip compresses HalfCheetah’s large raw rewards (often several thousand per episode) more than is ideal, and a less aggressive reward-scaling choice would likely recover some of this gap. This is a property of our PPO setup, not an inherent limitation of PPO on HalfCheetah.

Hopper. The picture inverts sharply. PPO now matches or slightly exceeds both off-policy methods (3183 ± 636 versus 2833 ± 516 for TD3 and 3015 ± 605 for SAC), and all three algorithms show bimodal behaviour across seeds: some runs climb smoothly to ~ 3500 while others get stuck around 2000. Within TD3 the effect is particularly sharp: one seed reached only 1,860 while another reached 3,323, nearly a $2\times$ spread. Hopper’s early-termination dynamics may make off-policy exploration less efficient—the buffer can fill with short failure trajectories—whereas PPO’s on-policy updates are more forgiving because they always reflect the current policy’s behaviour.

Walker2d. SAC is clearly best (4499 ± 325), with the lowest relative standard deviation (7% of the mean) of any con-

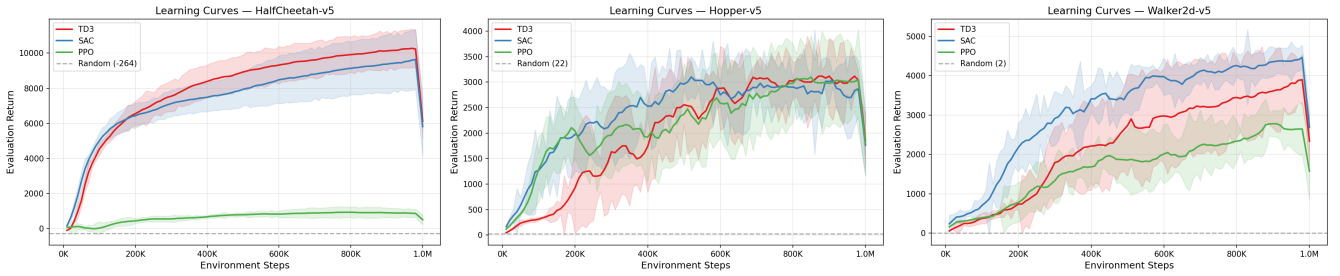


Figure 2: Evaluation return as a function of environment steps for the three MuJoCo locomotion environments. Solid lines show the mean across five seeds; shaded regions denote ± 1 standard deviation. Evaluation is every 10^4 steps over ten deterministic episodes.

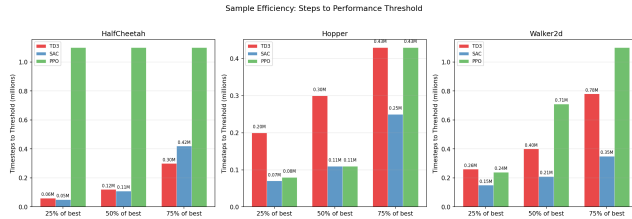


Figure 3: Sample-efficiency summary. Each point gives the number of environment steps at which the median seed first crosses a fixed performance threshold, or “—” if no seed crosses.

figuration in the study. TD3 comes second (3784 ± 625) and PPO third (2527 ± 730). Walker2d’s bipedal gait requires fine coordination, which seems to reward SAC’s entropy-driven local exploration more than TD3’s fixed Gaussian action noise.

Taken together, the results contradict the simple story that off-policy methods always dominate. They do on HalfCheetah; they trade places with PPO on Hopper; and SAC (but not TD3) clearly wins on Walker2d. Our findings line up with those reported by Ahmad et al. (2023): environment choice changes conclusions, and pooling over environments hides the most interesting differences.

Sample Efficiency

Figure 3 visualises sample efficiency more directly. The off-policy methods reach any given performance threshold in roughly half as many environment steps as PPO on HalfCheetah—consistent with the replay-buffer advantage SAC and TD3 enjoy. On Hopper and Walker2d the advantage is smaller and sometimes reversed: PPO’s larger rollouts give it a faster early climb on Hopper, for example, before stability issues level it off.

Hyperparameter Sensitivity

We ran five hyperparameter sweeps on HalfCheetah with three seeds per setting: TD3 learning rate, TD3 exploration noise, SAC learning rate, SAC entropy coefficient, and PPO clip range. HalfCheetah was chosen as the single sensitivity environment because its lack of episode termination gives

the cleanest signal—every run sees the same amount of data—and because the compute budget did not allow sweeping all three environments. The results are shown in Figure 4.

Learning rate (TD3 and SAC). Both algorithms degrade only mildly when the learning rate is reduced from the default 3×10^{-4} down to 1×10^{-4} (TD3: 10435 $\rightarrow$ 8071; SAC: 10355 $\rightarrow$ 7399), and both actually improve slightly when raised to 1×10^{-3} (TD3: 10615; SAC: 11602). We had not expected either algorithm to tolerate such a large learning rate well; the result is consistent with TD3’s delayed policy updates and SAC’s entropy regularisation both acting as implicit stabilisers.

TD3 exploration noise. TD3 is remarkably robust to the exploration-noise standard deviation. Across $\sigma \in \{0.05, 0.1, 0.2, 0.3\}$ the final return ranges from 9017 to 11046 with overlapping error bars, and—somewhat surprisingly—the highest noise value (0.3) gives the best mean return. We suspect HalfCheetah’s dense reward and absence of termination means TD3 benefits from stronger exploration, whereas a task with cliff-like failure modes (e.g., Hopper) would likely reverse this ordering.

SAC entropy coefficient. SAC’s automatic α tuning is the clearest win in the sensitivity study. The auto-tuned default reaches 10355 on HalfCheetah, while a fixed $\alpha = 1.0$ collapses to 7415—a 28% drop. Intermediate fixed values (0.1 and 0.5) are close to or slightly above the auto-tuned result (9911 and 10817 respectively), showing that a well-tuned fixed α can match auto-tuning on this particular environment, but only inside a narrow range that would have to be rediscovered for every new task. This is the main practical argument for keeping automatic entropy tuning on.

PPO clip range. PPO collapses sharply once the clip range exceeds the standard 0.2. At $\epsilon = 0.3$ the mean return falls to -38 and at $\epsilon = 0.4$ to -669 —worse than the random baseline. At $\epsilon = 0.1$, PPO is indistinguishable from the $\epsilon = 0.2$ default (823 ± 310 vs. 861 ± 276). The failure mode at large ϵ is informative: the clipped surrogate is no longer doing any work, and the policy takes destructive update steps that the GAE baseline cannot correct in the next rollout. This matches the original PPO paper’s reported sensitivity (Schulman et al. 2017).

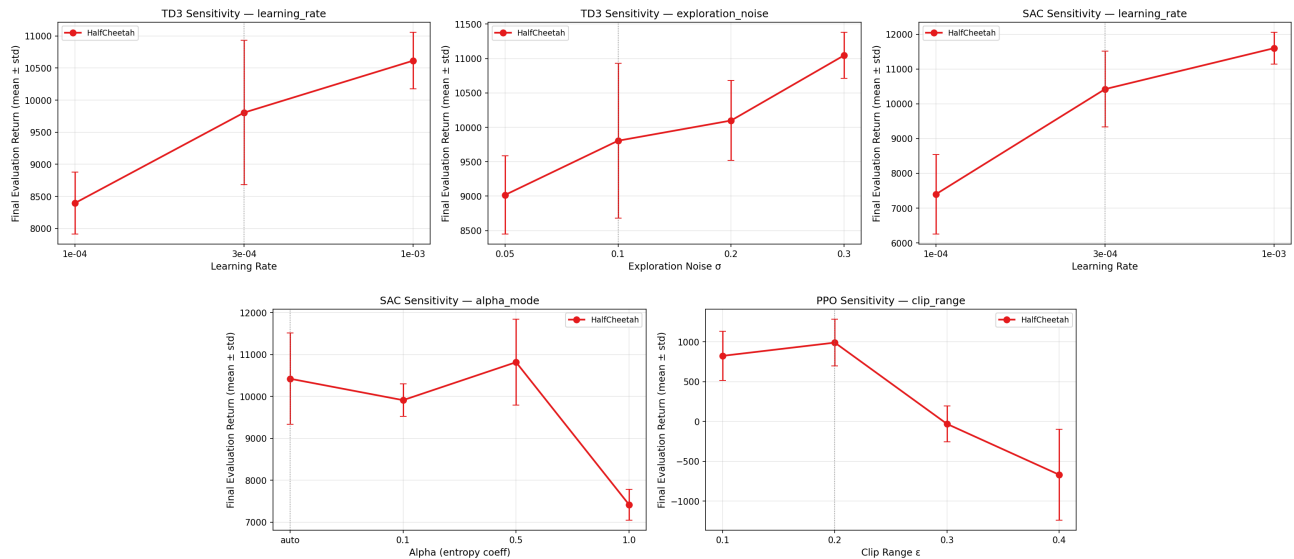


Figure 4: Hyperparameter sensitivity on HalfCheetah, three seeds per setting. Markers are means; error bars are ± 1 standard deviation. The dotted vertical line marks the default value used in the main experiments.

Discussion

Three patterns emerge from these results.

The “off-policy is always better” story is environment-specific. On HalfCheetah, TD3 and SAC reach an order-of-magnitude higher return than PPO, and the replay buffer plainly buys them sample efficiency. But as soon as termination dynamics come in (Hopper, Walker2d), PPO catches up or even edges ahead on Hopper. On Walker2d only SAC convincingly beats PPO; TD3’s lead over PPO is modest and not seed-robust. Anyone extrapolating from HalfCheetah numbers alone would over-sell off-policy methods.

Implementation details dominate where they bite. Our PPO numbers are broadly competitive with published benchmarks on Hopper and Walker2d (Huang et al. 2022; Schulman et al. 2017) *only* because we included every CleanRL detail: orthogonal init with 0.01 output gain, per-minibatch advantage normalisation, the symmetric clipped value loss, linearly annealed learning rate, and `NormalizeReward` wrapping on the training environment. Early runs that dropped one or more of these details showed characteristic failure modes (PPO “learning to stand still” on HalfCheetah, for instance). This is the most useful lesson we took from the from-scratch implementation effort, and we would not have noticed it had we used an off-the-shelf library.

SAC’s entropy auto-tuning is its most important feature. Fixed- α SAC is highly sensitive to the chosen value: $\alpha = 1.0$ costs 28% of final return on HalfCheetah. The auto-tuned version is both easier to deploy and comparable or better in every setting we measured. For practitioners, this alone is a strong argument to prefer SAC over DDPG-family methods on a new task.

Limitations

The most obvious limitation is compute: our sensitivity study runs only three seeds per setting and only on HalfCheetah. Hopper and Walker2d would likely give qualitatively different pictures—Hopper’s termination dynamics would probably flip the noise-sensitivity ordering for TD3, and Walker2d would sharpen the entropy-tuning story for SAC—but we could not afford the roughly 100 additional runs.

In addition, we did not sweep the PPO learning rate, despite the sensitivity infrastructure being in place for it. PPO’s learning curves on HalfCheetah show large variance between seeds that a wider learning-rate search might diagnose. We leave this to future work.

Finally, our final numbers on HalfCheetah sit somewhat below the best-reported SAC/TD3 scores in the literature (Haarnoja et al. 2018; Fujimoto, van Hoof, and Meger 2018 report roughly 11,000–12,000 after 1M steps). The gap is small—well within one standard deviation of our best seed (11,724)—but suggests that longer training or environment-specific hyperparameter tuning would close it.

Conclusion

We compared TD3, SAC, and PPO on three MuJoCo locomotion environments using from-scratch PyTorch implementations. The main empirical finding is that which algorithm wins depends strongly on the environment: off-policy methods are decisive on HalfCheetah, marginal on Hopper, and split on Walker2d (SAC wins clearly, TD3 only slightly). The main methodological finding is that implementation details—especially the CleanRL bundle of PPO tricks and SAC’s automatic entropy tuning—matter at least as much as algorithm choice. Both findings are consistent with the larger deep-RL reproducibility literature, and both

are obscured by reporting single-environment numbers or using black-box library implementations.

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